

Restaurant Online

Queuing System

Project Proposal

Course: 2190512 Application Development

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React Native

React.js

Flask

MongoDB



Background & Problems

Dining out is a daily activity, yet queue management remains unchanged – customers stand in long lines with no real-time information, wasting time and leaving frustrated.



Long Physical Queues

Customers abandon restaurants at peak hours, causing revenue loss



No Real-Time Visibility

No wait-time estimates lead to frustration & poor experience



Paper-Based Queue Lists

Manual lists cause errors, lost entries, and slow service



Overwhelmed Staff

Staff spend time managing walk-ins instead of hospitality



No Remote Joining

Customers can't join queues in advance – arrive too early or late



Fragmented Systems

No unified platform for both customers and staff

Scope of Application

Customer

✓ Web

✓ Mobile App

- Browse restaurants & see queue length
- Join queue remotely via web or mobile
- Track live position & wait time
- Receive alerts when table is ready
- View booking history

Restaurant Staff (Admin)

✓ Web

✓ Mobile App

- Login with Staff ID for admin access
- Manage full queue from dashboard
- Call next group with one tap
- Update table availability on the go
- Configure restaurant settings

Both platforms sync in real-time via Flask REST API + MongoDB

Features & Functionalities

Customer (Web + App)

1



Restaurant Discovery

Browse restaurants, see queue length & wait time

2



Queue Join

Select party size, join via web, app, or QR code

3



Live Queue Status

Real-time position tracker & countdown timer

4



Booking History

View upcoming and past queue entries

Staff (Web + App)

5



Staff ID Login (Admin)

Login with Staff ID + PIN for admin-level access

6



Queue Dashboard

Full queue list; call next party with one tap

7



Table Status Manager

Mark tables available/occupied, assign parties

8



Restaurant Settings

Set hours, queue capacity, turn-time estimates

Authentication & Access Control



Customer Login

Email / Social Sign-In

- ✓ Register with email & password
- ✓ Optional social login (Google)
- ✓ JWT token stored securely on device
- ✓ Access: browse, join queue, history
- ✓ Session persists across app restarts



Staff ID Login (Admin)

Staff ID + PIN / Password

- ✓ Login with unique Staff ID issued by restaurant
- ✓ PIN or password-based authentication
- ✓ JWT with admin-level role claims
- ✓ Access: queue mgmt, tables, settings
- ✓ Enforced session timeout for security

Security: All API endpoints protected by JWT – role claims determine customer vs. staff access level

Limitations



No Payment Integration

Payment or pre-ordering is out of scope; focus is queue management only



No GPS Tracking

App does not track customer location to auto-check-in on arrival



Notification Dependency

Push alerts require app installation and permission grant from the user



Internet Required

No offline mode; all queue data is live-synced with the server



Single Queue Per User

Customers can only be active in one restaurant queue at a time



No Live Chat

Staff-customer communication limited to automated status notifications

Technology Stack

Mobile Frontend



React Native + TypeScript + Expo

Android frontend implementation

Web Frontend



React Native + TypeScript + Expo

Shared component logic; responsive for both roles

Backend / API



Python + Flask

Lightweight REST API; fast to build and deploy

Database



MongoDB (NoSQL)

Flexible document model for queue & restaurant data

Authentication



JWT (JSON Web Tokens)

Stateless auth with role claims: customer vs. staff

Real-time Sync



Flask-SocketIO / Polling

Live queue position sync between server and all clients

Application Pages

CUSTOMER

WEB

APP



1

Restaurant Discovery / Browse

Search and browse restaurants. View current queue length and estimated wait time before joining.

WEB + APP



2

Queue Join & Live Status

Join remotely via web, app, or QR code. Select party size and track live position with a countdown timer.

WEB + APP



3

Booking History

Review all past and upcoming queue entries with timestamps, restaurant name, and party size.

WEB + APP

RESTAURANT STAFF (ADMIN)

WEB

APP



4

Queue Management Dashboard

Full live queue list with party details. Call the next group with one tap and manage the entire floor from a single view.

WEB + APP



5

Table Status Manager

Mark tables as available or occupied and assign arriving parties. Update table status on the fly from web or mobile.

WEB + APP



Thank You

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